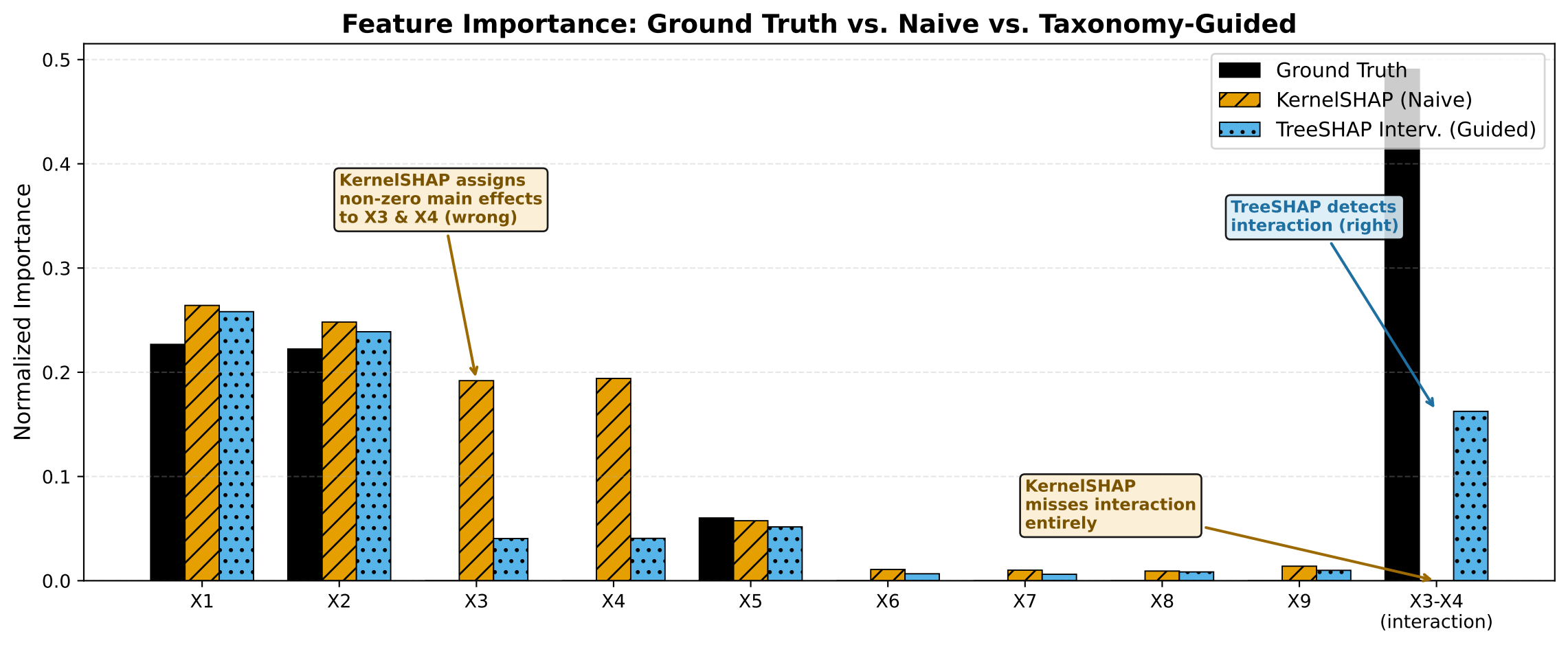
Feature Importance: Ground Truth vs. Naive vs. Taxonomy-Guided
Ground Truth
KernelSHAP (Naive)
TreeSHAP Interv. (Guided)
KernelSHAP assigns non-zero main effects to X3 & X4 (wrong)
TreeSHAP detects interaction (right)
KernelSHAP misses interaction entirely
Normalized Importance
0.0
0.1
0.2
0.3
0.4
0.5
X1
X2
X3
X4
X5
X6
X7
X8
X9
X3-X4 (interaction)